



Installation and Commissioning Instructions

Automation Marine
mtu NautIQ CoOperate

E532910/01E



A Rolls-Royce
solution

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Table of Contents

Preface

Intended Audience for This Document	v
Associated Documents	v
Book Conventions	vi

Chapter 1: Overview

Introduction	1-2
System Overview	1-2
mtu NautIQ CoOperate Connections	1-3
Installation Overview	1-4
Customer-Provided Components	1-5
Minimum Installation Requirements	1-6
Safety Considerations	1-7
Warnings and Cautions	1-7

Chapter 2: Planning Your Installation

Introduction	2-2
Installation Planning Worksheet	2-2
Locating Installation Procedures	2-5

Chapter 3: Mechanical Installation

Overview of the mtu NautIQ CoOperate Mechanical Installation	3-2
Installing the mtu NautIQ CoOperate Console on Your Vessel	3-3
Positioning the mtu NautIQ CoOperate UI Laptop on Your Vessel	3-5
Installing the Receiver/Antenna for the mtu NautIQ CoOperate Remote Helm	3-6
Installing the Optional Beacon and/or Antenna	3-7
Mounting Helm Controls	3-9

Chapter 4: Electrical Installation

Introduction	4-2
Connection Types Used in mtu NautIQ CoOperate	4-3
Ports Available on the mtu NautIQ CoOperate Console	4-3
Port Connectors	4-4
Minimum System Connection Requirements	4-5
Connecting Power to the mtu NautIQ CoOperate (Port A12)	4-6
Connecting the Propulsion System to the mtu NautIQ CoOperate	4-7
Connecting the Propulsion System Using CAN Bus (Port B10)	4-7
Connecting the Propulsion System Using an Analog Connection (Ports A3 & A4)	4-9
Connecting the Steering System to the mtu NautIQ CoOperate	4-12
Connecting the Steering System Using CAN Bus (Port B10)	4-12
Connecting the Steering System Using Analog Connections or Digital Relays (Ports A5 & A6)	4-14
Connecting the GPS/Compass to the mtu NautIQ CoOperate (Port B12)	4-17
Connecting the Vessel's System Stop to the mtu NautIQ CoOperate (Port A16)	4-18

Making Network Connections (Ports B1-B8)	4-19
Connecting the mtu NautIQ CoOperate Remote Helm to the mtu NautIQ CoOperate (Port B11).....	4-19
Connecting Marine Instruments to NMEA 2000 (Port B9)	4-20
Connecting a Bow or Stern Thruster to the mtu NautIQ CoOperate.....	4-23
Connecting the Bow or Stern Thruster Using the NMEA 2000 Network (Port B9)	4-23
Connecting the Bow or Stern Thruster with Discrete I/O (Port A7)	4-23
Connecting Bucket Control to the mtu NautIQ CoOperate	4-24
Connecting Bucket Control Using CAN Bus (Port B10).....	4-24
Connecting Bucket Control Using the NMEA 2000 Network (Port B9)	4-24
Connecting Serial Sensors Using NMEA 0183 Strings (Ports B13-B15).....	4-25
Connecting Auxiliary I/O	4-27
Connecting Discrete Outputs (Ports A1, A2, A8-A11, and A13 to A15).....	4-27
Discrete Inputs (Ports A9-A11).....	4-28
Auxiliary Analog Inputs.....	4-29

Chapter 5: Network Installation

Introduction	5-2
Types of Network Connections	5-3
Local Operation.....	5-3
Remote Operation.....	5-4
Making Network Connections	5-5
Connecting the Cameras to the Camera Ports (Port B1-B4)	5-6
Connecting the mtu NautIQ CoOperate UI to the UI/Comm Port (Port B5)	5-7
Connecting the mtu NautIQ CoOperate Internal Network Ports (B6-B7).....	5-7
Connecting the Furuno NavNet Radar to the Furuno Network Port (Port B8).....	5-8
Verifying Your Network Connections	5-10
Verifying the Local Network	5-10
Verifying the Remote Network	5-11

Chapter 6: Commissioning

Commissioning the mtu NautIQ CoOperate	6-1
Rolls Royce Power Systems AG mtu NautIQ CoOperate	
Commissioning Checklist.....	6-2
Dockside	6-2
Underway	6-2
Collaborative Following	6-3

Appendix A: Specifications

Appendix B: Ports on the mtu NautIQ CoOperate Connector Plate

Ports in Group A	B-1
Ports in Group B	B-9

Appendix C: Contact Person/Service Partner

Service	C-1
Local Support.....	C-1
24-h Hotline.....	C-1
Spare Parts Service	C-1

Preface

Welcome to the Rolls Royce Power Systems AG *Installation and Commissioning Instructions mtu NautIQ CoOperate*.

Intended Audience for This Document

The *Installation and Commissioning Instructions mtu NautIQ CoOperate* is intended for users with mechanical, electronic, and networking knowledge. The mtu NautIQ CoOperate must be installed by a qualified marine electronics technician. For electrical installations, the installer must comply with the NMEA 0400 Installation Standard. The installer must also comply with ISO/TR 10134:2020,

Small craft — Electrical devices — Established practices for the design, construction and installation of lightning-protection systems, for lightning protection.

The Commissioning procedure that takes place after installation will be performed by Rolls Royce Power Systems AG personnel. They will assist you in performing the sea trials at the end of the commissioning procedure.

Note: Only qualified personnel who are fully informed of the system safety hazards should install or operate this equipment.

Associated Documents

- *Operating Instructions mtu NautIQ CoOperate E532909/01E*

Book Conventions

This guide contains the following information highlights and cross-references:



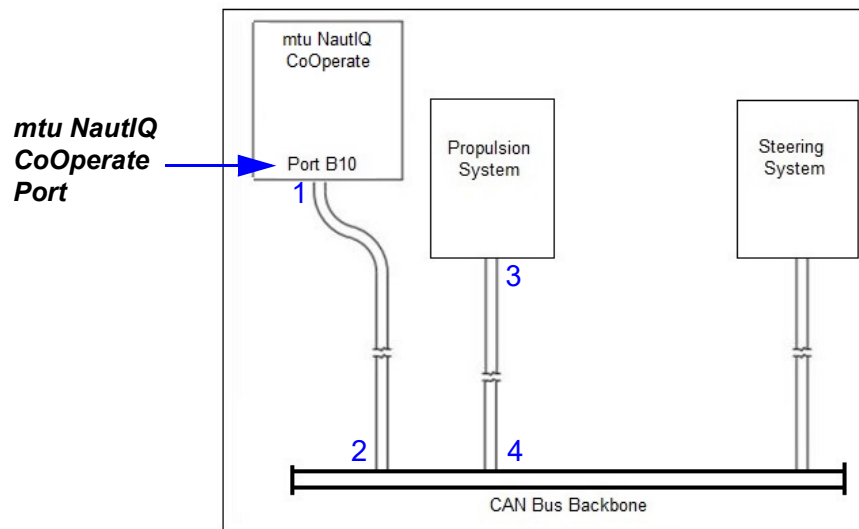
Warning: Warns the user regarding actions that may result in physical damage to the system or personal injury.



Caution: Cautions the user regarding actions that may result in operational issues or data loss.

Note: Identifies important points, helpful hints, special circumstances, or alternative methods.

- Cross references indicate the location of additional information regarding the chosen topic. References may include either headings on specific pages or entire chapters, for example, See “**Introduction**” on page 1-2.
- Buttons, field names, and selections are shown in **bold**.
- Blue boxes, text, and arrows highlight areas of interest on screenshots or images as shown:



Chapter 1:

Overview

Topics described in this chapter include:

Introduction	page 1-2
System Overview	page 1-2
mtu NautIQ CoOperate Connections	page 1-6
Installation Overview	page 1-4
Customer-Provided Components	page 1-5
Minimum Installation Requirements	page 1-6
Safety Considerations	page 1-7
Warnings and Cautions	page 1-7

Introduction

This guide describes how to plan for and install your mtu NautIQ CoOperate system.

System Overview

The mtu NautIQ CoOperate system includes three major components (see **Figure 1-1**):

- mtu NautIQ CoOperate Console
- mtu NautIQ CoOperate User Interface Laptop (mtu NautIQ CoOperate UI)
- mtu NautIQ CoOperate Remote Helm Controller (mtu NautIQ CoOperate Remote Helm)

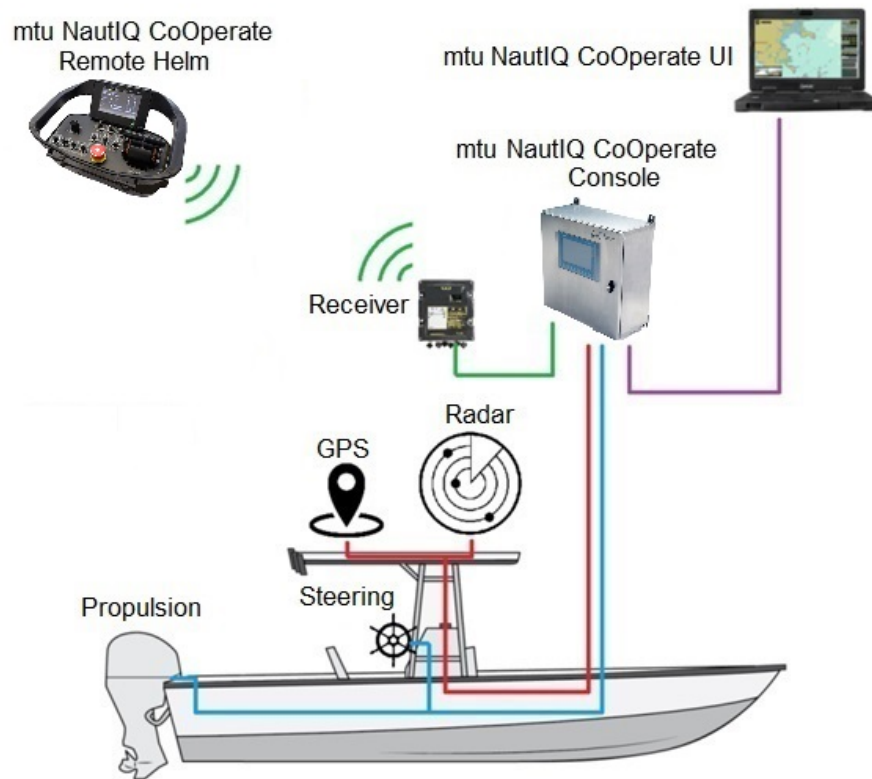
Figure 1-1: mtu NautIQ CoOperate System



mtu NautIQ CoOperate Connections

Figure 1-2 shows an overview of the mtu NautIQ CoOperate connections.

Figure 1-2: mtu NautIQ CoOperate Connections

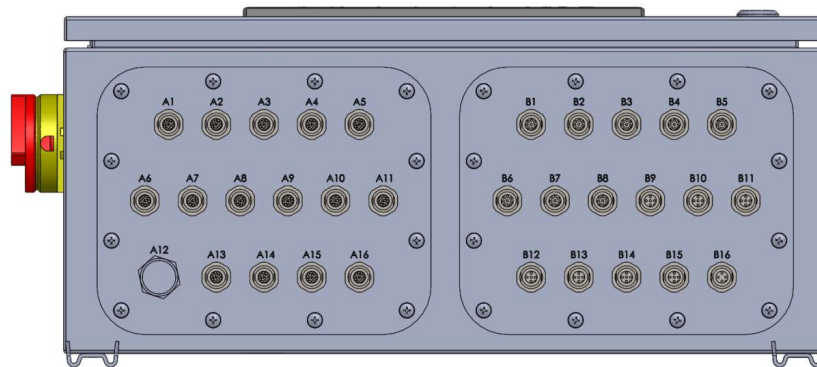


The mtu NautIQ CoOperate Console is mounted within your vessel. (The mtu NautIQ CoOperate Console is designed to be installed in a dry and environmentally protected space within the vessel. If there is no option to install within such a space, the mtu NautIQ CoOperate Console should be installed in a location that protects it from adverse weather conditions, water spray, and direct sunlight, and it should be mounted high enough to avoid any standing water.) The mtu NautIQ CoOperate UI connects to the Console via an Ethernet connection (either hardwired or over a wireless solution provided during installation). The mtu NautIQ CoOperate Remote Helm communicates to the mtu NautIQ CoOperate via a receiver that connects to the Console.

The vessel's steering and propulsion systems and all the vessel's onboard electronics, including the GPS, radar, and other auxiliary equipment all connect to the mtu NautIQ CoOperate Console.

Figure 1-3 shows the connector plate on the bottom of the mtu NautIQ CoOperate Console through which all connections are made.

Figure 1-3: Connection Panel on mtu NautIQ CoOperate Console



Refer to the *mtu NautIQ CoOperate User's Guide* for a full system overview.

Installation Overview

There are three main parts to the mtu NautIQ CoOperate installation:

- Mechanical Installation
- Electrical Installation
- Network Installation

Additionally, after installation, a Commissioning process is performed by a Rolls Royce Power Systems AG engineer or an authorized representative.